INES Ruhengeri - SWE 3513 - Individual Assignment 2

Test Log - pytest -q

```
.........                                                                [100%]
============================== warnings summary ===============================
tests/test_pipeline.py::test_api_health_and_predict
  C:\Users\kapor\OneDrive\Microsoft Copilot Chat Files\AI_A2_2527483\.venv\Lib\site-packages\fastapi\testclient.py:1:
StarletteDeprecationWarning: Using `httpx` with `starlette.testclient` is deprecated; install `httpx2` instead.
    from starlette.testclient import TestClient as TestClient  # noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
9 passed, 1 warning in 11.10s
```

Manual verification (19 October 2026)

- run_all.py --student-id ug2527483@ines.ac.rw --seed 7483: SUCCESS
  SHA-256 fingerprint b9a0067604ae61a7add8d7e174a95e588645899008b97feac17f468bb1b8aff2
  reproduced identically on a second full run.
- predict.py --json samples/valid.json: SUCCESS
  -> yield_tons 11.2163, high_risk 0, probability 0.12, cluster 0, code AI-A2-7483
- predict.py --json samples/invalid.json: rejected with exit code 1 and field-level errors.
- FastAPI: GET /health 200, GET /model-info 200, POST /predict valid 200,
  POST /predict out-of-range rainfall 422 (structured Pydantic error).